Replication package — "Selected Experience and Rare-Event Risk: Endogenous Observation, Subjective Transformation, and Reputation under Hidden Counterfactuals"

Replication README for the manuscript prepared for submission to the *Journal of Risk and Uncertainty* (JRU).

- **Manuscript:** *Selected Experience and Rare-Event Risk: Endogenous Observation, Subjective Transformation, and Reputation under Hidden Counterfactuals*
- **Author:** Tatsuki Onishi
- **Public repository:** https://github.com/bougtoir/pricing-the-rain
- **README prepared:** 2026-11-09, corresponding to the submitted analysis. (See `results/reproduction_manifest.json` for the exact output checksums.)

Overview

This package contains all code, archived raw inputs, derived datasets, and frozen numerical outputs needed to reproduce every table, figure, and in-text number in the manuscript. The study has two computational components:

1. **Generic rare-event simulation / model analysis** — Monte Carlo simulations of endogenous observation, subjective probability transformation, and reputation across four rarity regimes (`simulation/`).

2. **Real-weather known-zero benchmark** — archived ERA5 daily precipitation (1991–2020) for eight locations, fictional pseudo-events whose timing is drawn independently of rainfall, and model overlays of demand, willingness to pay (WTP), quotes, and reputation (`src/`, `data/`).

WTP, quote, and reputation quantities in the weather component are **model overlays calibrated on real weather**, not observed market data.

The full pipeline from archived inputs to manuscript outputs is driven by a single command (`make reproduce`; see *Instructions to replicators*). A full clean reproduction is CPU-bound during the Monte Carlo stages; expect on the order of **tens of minutes to a few hours** on a standard dual-core desktop (the pipeline was developed and validated on a 2-core, ~8 GiB RAM VM). Manuscript-facing outputs are found in `manuscript/`, `supplement/`, `tables/`, `analysis/out/`, and `results/manuscript_values.json`.

Data availability and provenance

☑ External (third-party) data are used in this paper, as detailed below; all other data are generated by the authors' code and are provided.

What is used

- **External data:** daily precipitation (ERA5 reanalysis, 1991-01-01 to 2020-12-31) for eight locations (Tucson, Cape Town, London, Mumbai, Brasilia, Darwin, Nairobi, Tokyo), retrieved through the Open-Meteo Historical Weather API, which serves the Copernicus C3S ERA5 product. Open-Meteo was the retrieval interface; the underlying dataset is ERA5.
- **Author-generated data:** the generic rare-event simulation outputs; fictional pseudo-event datasets (pseudo-event timing generated with a fixed seed, independent of rainfall); derived

analysis tables; and the machine-readable manuscript-value registry (`results/manuscript_values.json`).

- **Bibliographic metadata:** Crossref API records for every cited reference, archived in `data/raw/literature/`.

What is provided in the package

- The acquisition ledger for the 8 ERA5 series used in the analysis, `data/raw/acquisition_ledger.csv` (exact request URL, retrieval UTC timestamp, bytes, SHA-256, license note, completeness check), and the ledger of 8 ERA5-Land responses that contained only null precipitation and are excluded from all analysis (`data/raw/invalid_acquisition_ledger.csv`).
- All author-generated results needed to re-render the manuscript (`results/*.csv*`, `tables/*.csv*`, *aggregated* `*analysis/*.csv`) and `results/reproduction_manifest.json`, which holds the SHA-256 of the derived daily weather file so that a re-acquired series can be checked.

What is not redistributed — the files listed in `config/public_mirror_exclusions.txt`:

- the ERA5 daily precipitation snapshots (`data/raw/weather/`) and the daily files derived from them (`data/processed/daily_weather_features.csv.gz`, `analysis/pseudo_events.csv`). The authors do not redistribute ERA5; replicators obtain it from the public Open-Meteo archive API, and the pipeline regenerates the derived files;
- provenance-only journal guideline pages and two literature full texts (copyright or site terms). None of these is an analytical input.

`run_all.py` reports absent listed files as not redistributed and verifies them by size and SHA-256 when present.

Re-acquisition: `make reproduce` runs `src/acquire_weather.py` when the ERA5 snapshots are absent. This requires internet access, and the script never overwrites an existing snapshot. The script writes the ledger rows for the snapshots it retrieves. Open-Meteo responses carry request metadata (for example, generation time), so file bytes and SHA-256 differ from the authors' snapshots. The daily `time` and `precipitation_sum` arrays were identical on re-acquisition on 2026-11-11. The derived `data/processed/daily_weather_features.csv.gz` and all downstream outputs then match `results/reproduction_manifest.json`. `make acquire` also re-fetches the Crossref records.

Statement about rights

☑ The author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript: ERA5 is open Copernicus C3S data accessed through the Open-Meteo public archive API (attribution required), and Crossref metadata is public.

☑ ERA5 data are not redistributed in this package. Only acquisition ledgers, aggregated results and the retrieval code are provided. The Crossref records are public bibliographic metadata. The author-generated data are the author's own work.

License for data

- **Author-generated / simulated data:** MIT License (see `LICENSE`), same as the repository license for this package.
- **Upstream ERA5 data:** not redistributed; Copernicus C3S terms apply to re-acquired data (attribution to the Copernicus Climate Change Service).
- **Crossref metadata:** public bibliographic metadata.

Summary of availability

All author-generated data are provided. ERA5 is obtained from the public Open-Meteo archive API on the first `make reproduce`, and that is the only networked step besides installing the Python packages. After that the pipeline runs offline.

Details on each data source

| Data source / dataset | Role | Source/provider | Files used | Format | Included | Acquisition route | Citation / provenance | |---|---|---|---|---|---|---|---| | ERA5 daily precipitation (8 locations) | Real-weather benchmark input | Copernicus C3S ERA5 via Open-Meteo Historical Weather API | `data/raw/weather/*_era5_1991_2020.json` | *JSON* | *No (re-acquired)* | *`src/acquire_weather.py`; ledger `data/raw/acquisition_ledger.csv`* | *Hersbach et al. (2020), ERA5; see References* | | *ERA5-Land responses (8 locations)* | *Not used — all-null precipitation* | *Same interface, `model=era5_land`* | *`data/raw/weather/*_era5_land_1991_2020.json` | JSON | No (ledger only; not used) | Same; ledger `data/raw/invalid_acquisition_ledger.csv` | Acquisition audit trail | | Crossref records (28 works) | Reference metadata | Crossref REST API | `data/raw/literature/crossref_*.json` | *JSON* | *Yes* | *`src/acquire_literature.py`; ledger `data/raw/literature_acquisition_ledger.csv`* | *Per-DOI* | | *Daily weather features* | *Derived benchmark input* | *Generated* | *`data/processed/daily_weather_features.csv.gz`* | *CSV.gz* | *No (regenerated)* | *`src/weather_analysis.py`* | *Derived* | | *Generic simulation results* | *Model evidence* | *Generated* | *`results/general_*.csv`, `results/general_summary.csv`* | *CSV/CSV.gz* | *Yes* | *`simulation/general_rare_event.py`* | *Derived* | | *Principal/sensitivity simulation results* | *Benchmark evidence* | *Generated* | *`results/simulation_*.csv*` | CSV/CSV.gz | Yes | `simulation/run_simulation.py`, `analysis/robustness.py` | Derived | | Manuscript-value registry |

In-text numbers | Generated | `results/manuscript_values.json` | JSON | Yes |

`analysis/build_manuscript_assets.py` | Derived |

ERA5 snapshots — variables: daily `precipitation_sum` (mm); span 1991–2020 (10,957 days per location); grid cell at each location's coordinates as recorded in the request URLs in the acquisition ledger; transformations in `src/weather_analysis.py` (daily features, hazard and threshold variables, pseudo-event sampling with seed 20260924).

Computational requirements

Software

- Python 3.10 (validated on Python 3.10.12, Ubuntu 22.04)
- GNU Make; shell
- LibreOffice (`libreoffice`/`soffice`) — required only for DOCX→PDF rendering of manuscript assets, not for any numerical result
- Python packages pinned exactly in `requirements.txt` (numpy 2.2.6, pandas 2.3.3, scipy 1.15.3, statsmodels 0.14.5, matplotlib 3.10.9, seaborn 0.13.2, python-docx 1.2.0, python-pptx 1.0.2, PyYAML 6.0.2, reportlab 4.4.4, requests 2.34.2, CairoSVG 2.8.2, latex2mathml 3.77.0, docx-equation 0.3.0, pytest 8.3.4)
- Single setup command: `make reproduce` (creates `.venv`, installs the pinned requirements, runs the full pipeline)

Controlled randomness

| Use | Seed | Location | |---|---|---| | Principal Monte Carlo simulation (incl. weather-selection and per-cell draws via `np.random.SeedSequence` spawn) | 20260924 | `config/simulation.yml` | | Generic rare-event simulation (all regime × rule cells and equivalent-efficacy cells via

`SeedSequence` spawn) | 20261012 | `config/general_simulation.yml` | | Robustness/sensitivity scenarios | 20260925 | `analysis/robustness.py` (`SENSITIVITY_SEED`) | | Price–success permutation draws | 20260924 | `analysis/robustness.py` (`PERMUTATION_SEED`) | | Fictional pseudo-event / random-drought selection on real weather | 20260924 | `src/weather_analysis.py` (line ~192) |

All seeds are fixed constants; `np.random.SeedSequence.spawn` gives each replication cell an independent but deterministic sub-stream. No parallel execution is used; reruns on the pinned environment reproduce the machine-readable numerical outputs deterministically. DOCX/PPTX/PDF containers may differ in internal timestamps; all manuscript numbers come from deterministic JSON/CSV outputs.

Memory, runtime, storage

- **Runtime:** CPU-bound in the Monte Carlo stages; tens of minutes to a few hours for `make reproduce` on a 2-core desktop (validated on 2 vCPU / 7.8 GiB; see `docs/REPRODUCIBILITY.md`). `make manuscript` (rebuild only figures/tables/manuscript from existing outputs) takes minutes.
- **RAM:** < 4 GiB sufficient (validated at ~8 GiB).
- **Storage:** working tree ≈ 0.5 GiB excluding `.git`; the virtual environment adds ~1 GiB.

Description of programs

Canonical entry point: **`run_all.py`** (driven by `make reproduce`), which executes this ordered pipeline:

| Script | Purpose | Main outputs | |---|---|---| | `src/acquire_weather.py` | Optional networked ERA5 acquisition (skips existing snapshots) | `data/raw/weather/*.json`, *acquisition ledger* | |

`src/acquire_literature.py` | *Optional Crossref metadata acquisition* | *`data/raw/literature/`*.json`, literature ledger | | `src/model_diagram.py` | Framework diagram | `analysis/out/` | | `src/weather_analysis.py` | Weather features + pseudo-events on archived ERA5 | `data/processed/`, pseudo-event results | | `simulation/run_simulation.py` | Principal Monte Carlo models M1–M6 | `results/simulation_` | | *`simulation/general_rare_event.py`* | *Generic 4-regime × 2-rule simulation + equivalent efficacy* | *`results/general_`* | | `analysis/lookahead_correction_diff.py` | Pre- vs post-correction comparison of generic outputs (archive in `results/archive/pre_lookahead_fix_837387d/`) | `results/lookahead_correction_diff_*.csv*` | | *`analysis/robustness.py`* | *Sensitivity, permutation falsification, convergence* | *`results/simulation_sensitivity_`*, `analysis/*` | | `analysis/build_manuscript_assets.py` | Figures, tables, `results/manuscript_values.json` | `analysis/out/`, `tables/`, `results/` | | `manuscript/build_manuscript.py` | Manuscript/supplement/cover-letter DOCX+PDF, submission ZIP | `manuscript/`, `supplement/` | | `analysis/final_audit.py` | Statistical, traceability, and package audit | `results/final_audit.json` | | `run_all.py` (`write_manifest`) | SHA-256 manifest of outputs | `results/reproduction_manifest.json` |

License for code

MIT License — see `LICENSE` (Copyright (c) 2026 Tatsuki Onishi).

Instructions to replicators

1. Clone or download this repository at the submitted state (public mirror: `git clone https://github.com/bougtoir/pricing-the-rain`).

2. From the project root run `make reproduce`. This creates `.venv`, installs pinned dependencies, retrieves the ERA5 series from Open-Meteo if they are absent, verifies every raw input that is present against the acquisition ledgers (size + SHA-256), and runs the full pipeline: weather features → pseudo-events → simulations → robustness → figures/tables → `manuscript_values.json` → manuscript assets → final audit. Expected runtime: tens of minutes to a few hours; successful completion ends with the final audit passing.
3. (Optional) `make acquire` also re-fetches Crossref sources that are not already archived (requires internet).
4. `make test` runs the invariant and package test suite (33 tests). Run it after `make reproduce`, because some tests read regenerated daily files.
5. Outputs: manuscript assets in `manuscript/` and `supplement/`, tables in `tables/`, figures in `analysis/out/`, all in-text numbers in `results/manuscript_values.json`, and integrity hashes in `results/reproduction_manifest.json`.

Detailed program flow

acquisition (optional) → raw-ledger validation → weather features + pseudo-events → principal simulation → generic simulation → robustness/sensitivity/permutation → manuscript values, figures, tables → DOCX/PDF package → final audit → manifest + tests. Ordering constraints: simulation and benchmark outputs must exist before `build_manuscript_assets.py`; `manuscript_values.json` must exist before `build_manuscript.py` (it resolves every `{{TOKEN}}` numeric placeholder in the manuscript template — an unresolved token aborts the build).

Table / figure / in-text reproduction map

All four main tables (Tables 1–4), all supplementary tables (S1–S10), all main and supplementary figures, and every in-text numerical claim are reproduced by the pipeline above:

- Tables: `tables/table1–table6_*.csv` and `tables/table_s1–table_s10_*.csv`, editable DOCX `tables/main_tables_editable.docx`, `tables/supplementary_tables_editable.docx`, generated by `analysis/build_manuscript_assets.py` from `results/*.csv`.
- Figures: `analysis/out/fig*.png` plus editable `fig_*editable.pptx`, generated by `analysis/build_manuscript_assets.py` (and `src/model_diagram.py` for the framework diagram).
- In-text numbers: every claim is a `{{TOKEN}}` in `manuscript/manuscript_template.md` / `supplement/supplement_template.md` resolved from `results/manuscript_values.json` (and two model-theoretic values computed from `config/general_simulation.yml`); see `docs/JRU_NUMERIC_HARDCODE_AUDIT.md` and `docs/JRU_RESULT_TRACEABILITY.md` for the claim → source map.

References

- Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730), 1999–2049. https://doi.org/10.1002/qj.3803 (data via Copernicus Climate Change Service, accessed through the Open-Meteo Historical Weather API, https://open-meteo.com/)
- Crossref REST API, https://api.crossref.org/ — bibliographic metadata.
- Repository: https://github.com/bougtoir/pricing-the-rain

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